RAHUL SAHU

B.Tech Electrical & Electronics Engineering | Nawalparasi (West), Nepal
(+91) 7007696657 | sahurahuloc@gmail.com | linkedin.com/in/rahul-sahu-eee | github.com/RahulSahu1221

SUMMARY

Final-year B.Tech Electrical and Electronics Engineering student (CGPA 9.13/10) with hands-on experience in utility distribution maintenance, EV systems and industrial IoT. Seeking a 4-month internship from December 2026 in automation, power systems or electrical design.

EDUCATION

B.Tech, Electrical and Electronics Engineering – CGPA 9.13/10 **2023 – 2027**
Aditya College of Engineering & Technology (Autonomous), Surampalem, Andhra Pradesh

Higher Secondary (Technical & Vocational – Electrical Engineering) – GPA 3.43/4 **2020 – 2022**
Shree Pashupati Secondary School, Rupendehi, Nepal

Secondary (Technical Stream with Science) – GPA 3.92/4 **2016 – 2020**
Tribhuvan Secondary School, Nawalparasi (West), Nepal

EXPERIENCE

Embedded & IoT Intern – Emertxe **Aug 2026 – Sep 2026**
Edge AI Based Smart EV Charging Station Optimizer

- Built a simulated three-bay EV charging station on ESP32 (Wokwi) with on-device Edge AI predicting arrival demand and session duration.
- Implemented a priority-based algorithm to allow, throttle or defer charging per bay in real time, with telemetry and control through ThingsBoard over MQTT.

IoT Intern – Emertxe **Mar 2026 – Apr 2026**
Smart Factory IoT Monitoring System

- Built a smart factory simulation on Arduino Uno and PICSimLab with two embedded nodes monitoring production and warehouse conditions.
- Published real-time data to ThingsBoard Cloud via MQTT; implemented remote device control (RPC) and live dashboards.

On-the-Job Trainee – Nepal Electricity Authority, Bharatpur Distribution Centre **Jun 2022 – Dec 2022**

- Supported distribution maintenance and fault restoration; assisted in service cable replacement, energy meter installation and HT line patrol.
- Gained exposure to substation operations.

PROJECTS

Transformer Life Gauge (MATLAB; simulation-validated concept)

- Designed a low-cost IoT retrofit concept that estimates a transformer's remaining insulation life from load current and temperature using the IEEE C57.91 thermal aging model.
- Validated the model in MATLAB as an offline simulation and a live dashboard showing remaining life (%) and safe overload capacity.
- Aimed at mid-size transformers as a low-cost alternative to fiber-optic monitoring, using off-the-shelf IoT hardware.

SKILLS

Electrical	Circuit Analysis, Power Systems, Power Electronics, Control Systems
Design & Simulation	MATLAB, Simulink, Proteus 8 Professional, DIALux Evo, AutoCAD, Multisim, Ultiboard, PICSimLab, Wokwi
Embedded & IoT	Arduino, ESP32, STM32, MQTT, ThingsBoard
Practical	Electrical repair of motors, fans and household equipment
Languages	English, Hindi, Nepali, Bhojpuri (native); Japanese – JLPT N5 passed, studying N4

CERTIFICATIONS

- Industry 4.0 / Industrial IoT – NPTEL (2026) | PCB Design (Multisim, Ultiboard) – Alison
- MATLAB Onramp and Simulink – MathWorks | JLPT N5 – Japan Foundation (Dec 2025)

ACHIEVEMENTS

- **2nd Prize, Poster Presentation** – VEDA 2K25 National Level Student Symposium (Sep 2025)
- **Academic rank:** 2nd in Semesters 1–4 and 1st in Semester 5 (B.Tech); 3rd in Secondary level (2020)

LANGUAGES

English — Professional Working Proficiency | Hindi — Fluent | Nepali — Fluent | Bhojpuri — Native | Japanese — JLPT N5